

3(a)(i)	work done per unit time	B1
3(a)(ii)	$W = Fs$	B1
	$P = Fs / t$ <u>and</u> (so) $P = Fv$	B1
3(b)	$(F \Rightarrow) 130 \times 10^3 / 25 = 5200 \text{ (N)}$	A1
3(c)(i)	(component of weight $\Rightarrow mg \sin \theta$)	A1
3(c)(ii)	$F \text{ (along slope due to weight)} = 36\,000 \times 9.81 \times \sin 1.4^\circ$ $(= 8600 \text{ N})$	C1
	$(\text{total}) F = 5200 + 36\,000 \times 9.81 \times \sin 1.4^\circ$ $(= 13\,800 \text{ N})$	C1
	$P = 13\,800 \times 25$ $= 350 \times 10^3 \text{ (W)}$ $= 350 \text{ kW}$	A1